

Fetal Head Ultrasound Recognition

Deep Learning Final Project: Automated Biometry

Automating prenatal measurements through direct ellipse regression

Introduction and Motivation

Enhancing prenatal care accuracy through deep learning automation.

Improving Prenatal Care Efficiency

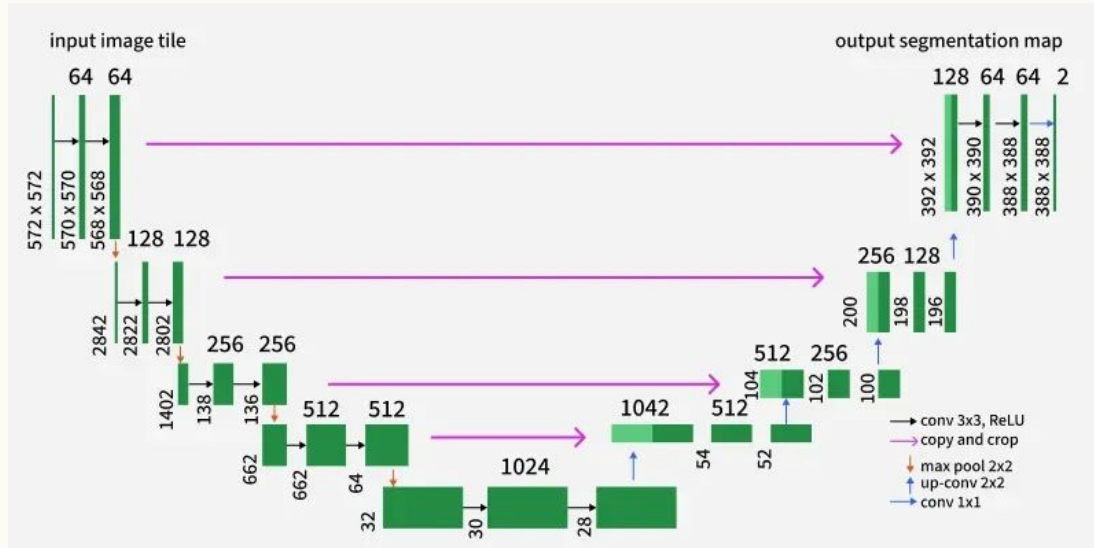
Manual Bottlenecks

Routine scans require sonographers to manually trace the fetal skull. This process is time-consuming and relies heavily on operator experience.

The Precision Gap

Significant inter-operator variability leads to inconsistent clinical decisions. AI minimizes human error and provides objective metrics.

Related work: U-Net approach



Foundation: Wang's Research

Wang utilizes a **U-Net architecture** with a MobileNetV2 encoder pre-trained on ImageNet.

- Outputs pixel-wise probability maps
- Binary masks generated via thresholding
- Ellipse derived via Hough Transform

Depthwise conv block: *Input* → *Conv2d* → *BatchNorm* → *ReLU* → *Conv2d* → *BatchNorm* → *ReLU* → *Output*

Dataset and Preprocessing

Preparing medical imagery for high-precision regression tasks.

The HC18 Ultrasound Dataset

Dataset Characteristics

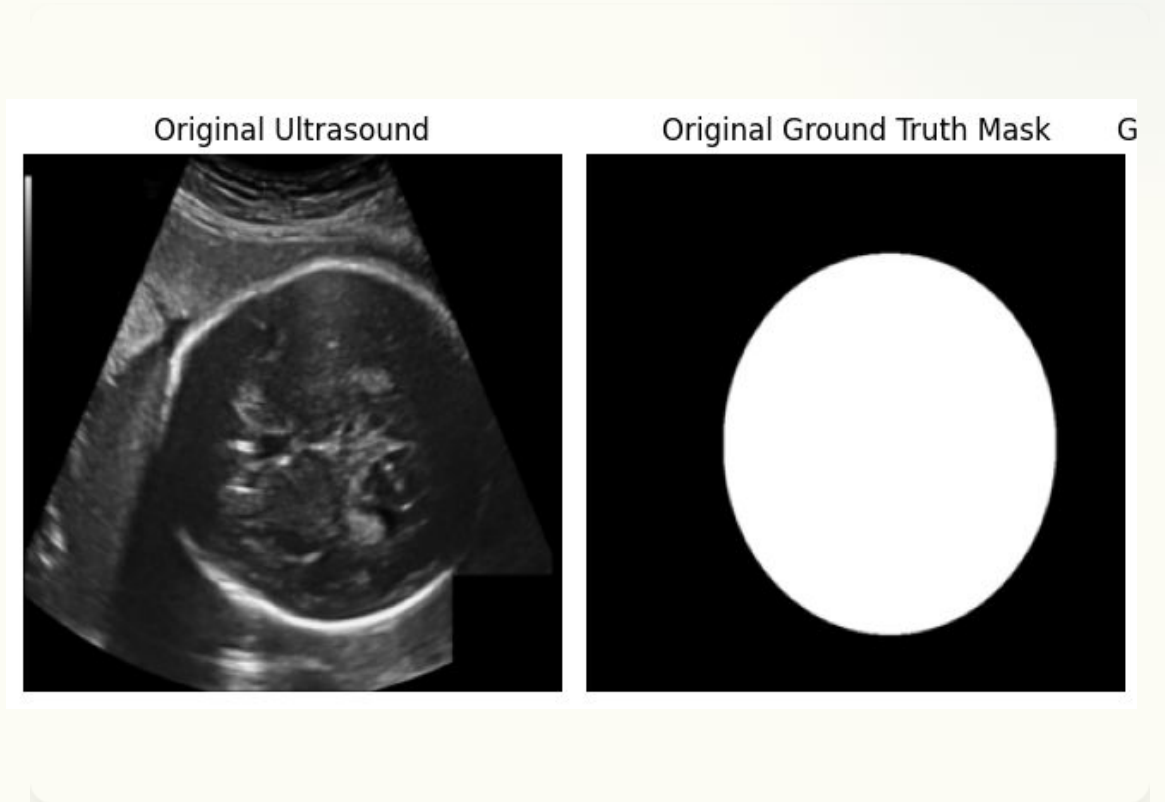
- 799 training and 200 testing images
- Ground truth segmentation masks (ellipse that best fits)

Our Dataset

- Masks converted to 6-parameters (center, axes, rotation)
- OpenCV functions were used for extracting the labels

Training Split

- 80% train → 639 samples
- 20% train → 160 samples



Regression Task Normalization

Problems:

- $(c_x, c_y), (d_{\max}, d_{\min})$ IN PIXEL UNITS!
- $\theta = 0^\circ$ equal to $\theta = 360^\circ$ BUT NUMERICALLY FAR APART!

To improve numerical stability and optimization convergence, target values are normalized to a smaller range.

$$\text{new target} = (c_x/w, c_y/h), (d_{\max}/w, d_{\min}/h), (\sin 2\theta, \cos 2\theta)$$

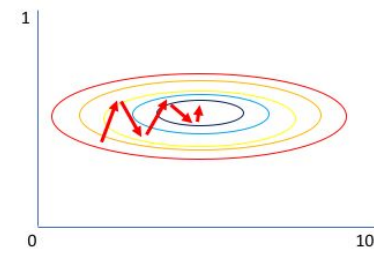
✓ Symmetry Handling:

Using $\sin(2\theta)$ accounts for 180° rotational symmetry.

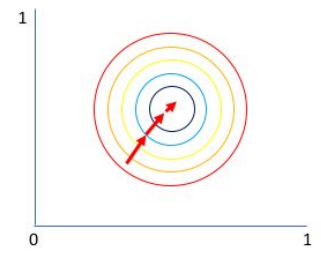
✓ Continuity:

Sine/Cosine encoding avoids boundary discontinuities (0° vs 360°).

Why normalize?



Gradient of larger parameter dominates the update



Both parameters can be updated in equal proportions

Methodology: MobileNetV3 Regression

Architecture Shift

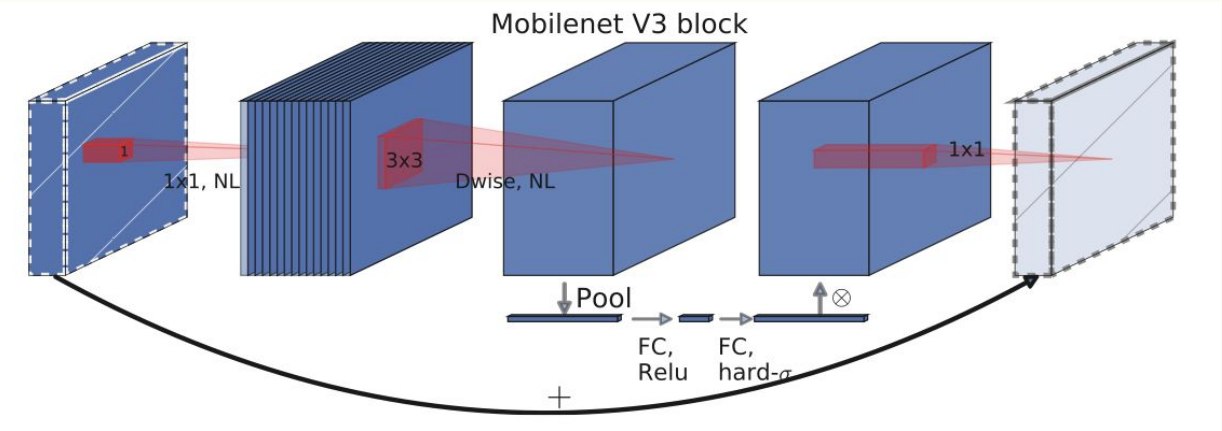
From pixel **classification** to direct parameter **regression**.

MobileNetV3

Input	Operator	exp size	#out	SE	NL	s
$224^2 \times 3$	conv2d, 3x3	-	16	-	HS	2
$112^2 \times 16$	bneck, 3x3	16	16	✓	RE	2
$56^2 \times 16$	bneck, 3x3	72	24	-	RE	2
$28^2 \times 24$	bneck, 3x3	88	24	-	RE	1
$28^2 \times 24$	bneck, 5x5	96	40	✓	HS	2
$14^2 \times 40$	bneck, 5x5	240	40	✓	HS	1
$14^2 \times 40$	bneck, 5x5	240	40	✓	HS	1
$14^2 \times 40$	bneck, 5x5	120	48	✓	HS	1
$14^2 \times 48$	bneck, 5x5	144	48	✓	HS	1
$14^2 \times 48$	bneck, 5x5	288	96	✓	HS	2
$7^2 \times 96$	bneck, 5x5	576	96	✓	HS	1
$7^2 \times 96$	bneck, 5x5	576	96	✓	HS	1
$7^2 \times 96$	conv2d, 1x1	-	576	✓	HS	1
$7^2 \times 576$	pool, 7x7	-	-	-	-	1
$1^2 \times 576$	conv2d 1x1, NBN	-	1024	-	HS	1
$1^2 \times 1024$	conv2d 1x1, NBN	-	k	-	-	1

➡ 11 bottleneck blocks

MobileNetV3-Small



15 bottleneck blocks

Input	Operator	exp size	#out	SE	NL	s
$224^2 \times 3$	conv2d	-	16	-	HS	2
$112^2 \times 16$	bneck, 3x3	16	16	-	RE	1
$112^2 \times 16$	bneck, 3x3	64	24	-	RE	2
$56^2 \times 24$	bneck, 3x3	72	24	-	RE	1
$56^2 \times 24$	bneck, 5x5	72	40	✓	RE	2
$28^2 \times 40$	bneck, 5x5	120	40	✓	RE	1
$28^2 \times 40$	bneck, 5x5	120	40	✓	RE	1
$28^2 \times 40$	bneck, 3x3	240	80	-	HS	2
$14^2 \times 80$	bneck, 3x3	200	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	184	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	184	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	480	112	✓	HS	1
$14^2 \times 112$	bneck, 3x3	672	112	✓	HS	1
$14^2 \times 112$	bneck, 5x5	672	160	✓	HS	2
$7^2 \times 160$	bneck, 5x5	960	160	✓	HS	1
$7^2 \times 160$	bneck, 5x5	960	160	✓	HS	1
$7^2 \times 160$	conv2d, 1x1	-	960	-	HS	1
$7^2 \times 960$	pool, 7x7	-	-	-	-	1
$1^2 \times 960$	conv2d 1x1, NBN	-	1280	-	HS	1
$1^2 \times 1280$	conv2d 1x1, NBN	-	k	-	-	1

MobileNetV3-Large

Methodology: Baseline model

Architecture:

- MobileNetV3-Small
- Entire backbone freezed

Training configuration:

- Adam optimizer
- 0.001 learning rate
- 15 epochs

Custom classifier

- FC layer compressor
- Hardwish and dropout of 20%
- Linear classifier of 6 values

Experiments and evaluated configurations



Fine-Tuning

Selective unfreezing of deep layers to adapt pre-trained ImageNet weights to ultrasound-specific features.



Data Augmentation

Simulating sonographer variability through rotation and normalization to prevent model overfitting.



Model Scaling

Leveraging MobileNetV3-Large to extract richer high-level abstract representations for improved biometry.

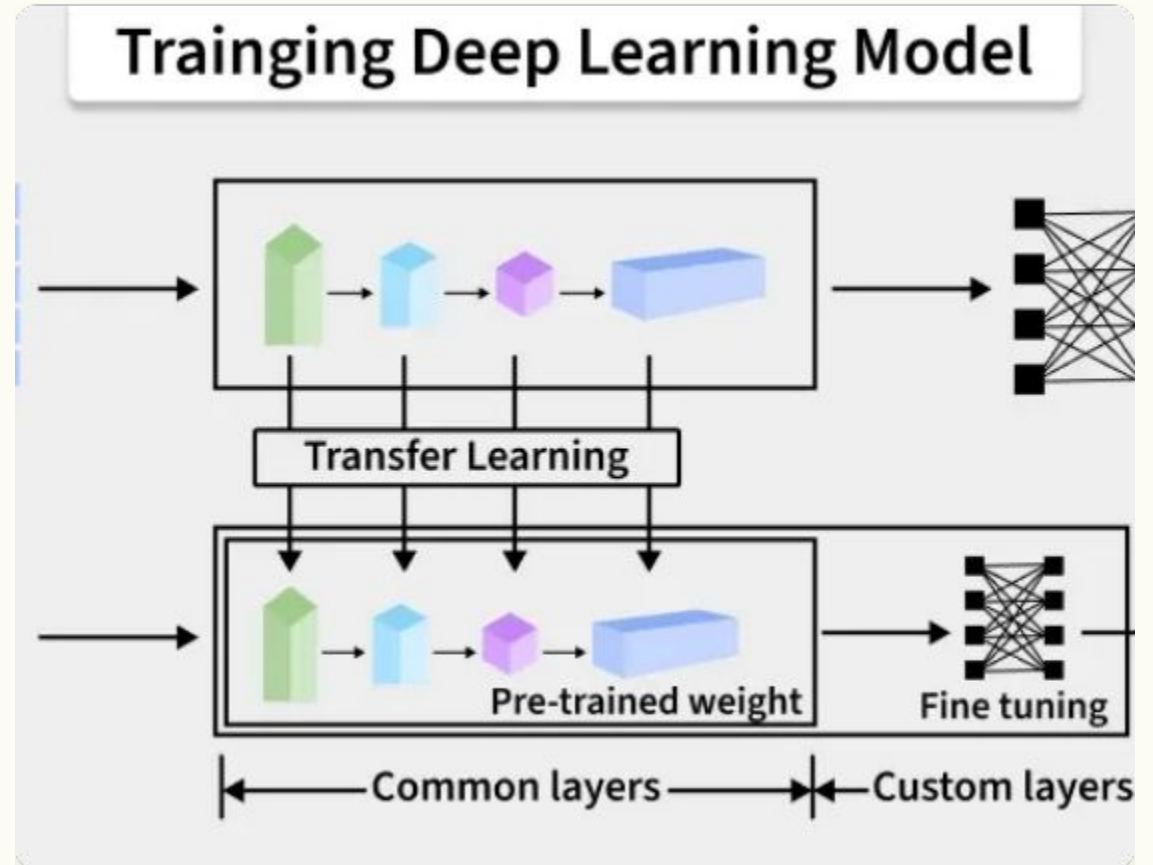
Fine-Tuning: Depth Layer Adaptation

Moderate Fine-Tuning

Unfreezing only the last 2 layers (Inverted Residual + Final Conv). Focuses on aggregating high-level embeddings for the custom classifier.

Aggressive Fine-Tuning

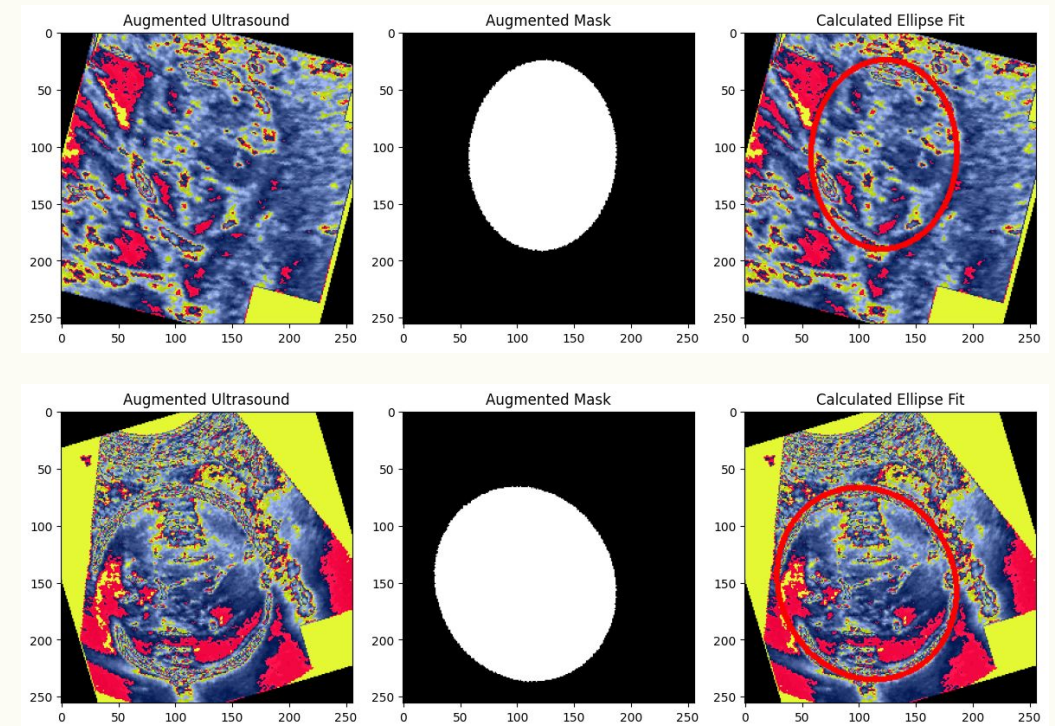
Unfreezing the last 6 layers. Adapts the network to the specific geometric curvature of the cranium and medical-grade noise patterns.



Data Augmentation

- **Random Rotation:** Images rotated between -20° and 20° to simulate various probe angles.
- **Synchronized Transform:** The exact same rotation matrix is applied to both the ultrasound image and its ground truth mask.
- **Distribution Matching:** Images are resized and normalized to align with the MobileNetV3 pre-trained distribution.
- **Color Jitter:** change the brightness and contrast of the images.

Goal: Create a model robust to clinical environmental noise and diverse scanning perspectives.



MobileNetV3-Large

While V3-Small provides efficiency, the **Large** backbone offers superior biometry precision.

- **Increased Depth:** More Inverted Residual blocks for hierarchical feature extraction.
- **Wider Channels:** Captures finer semantic details in grainy ultrasound textures.
- **Resource Usage:** Designed for high-resource environments where accuracy outweighs latency.

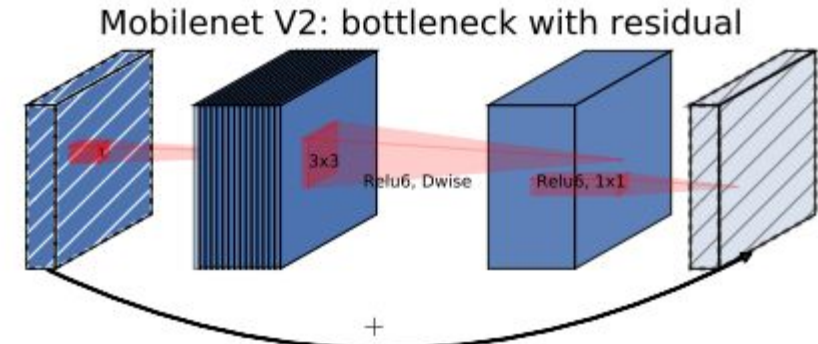


Figure 3. MobileNetV2 [39] layer (Inverted Residual and Linear Bottleneck). Each block consists of narrow input and output (bottleneck), which don't have nonlinearity, followed by expansion to a much higher-dimensional space and projection to the output. The residual connects bottleneck (rather than expansion).

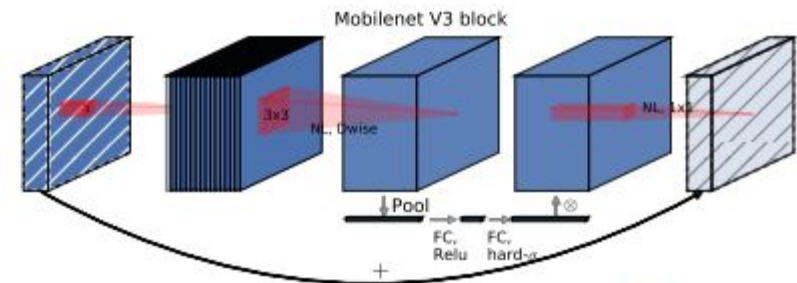
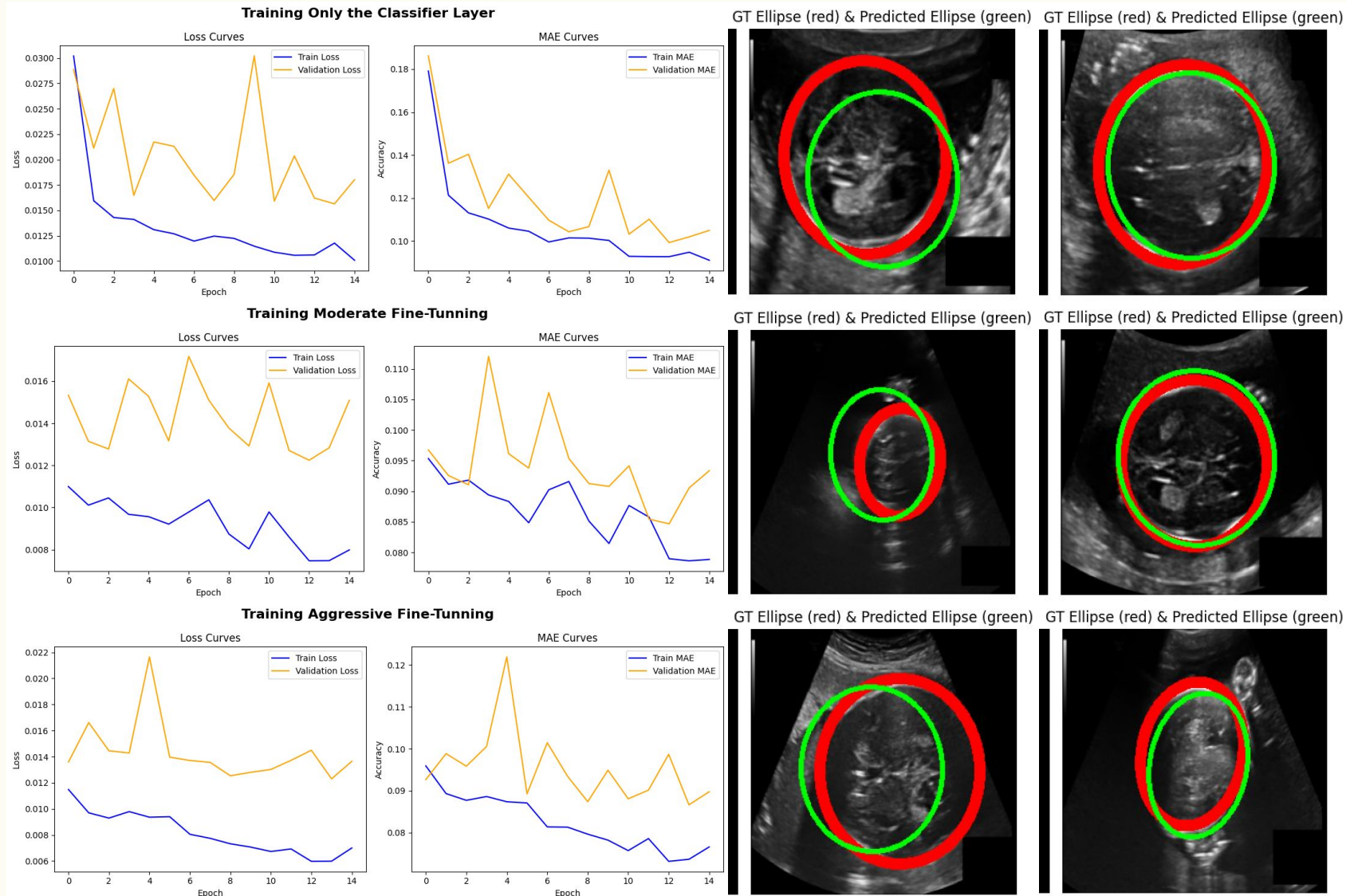
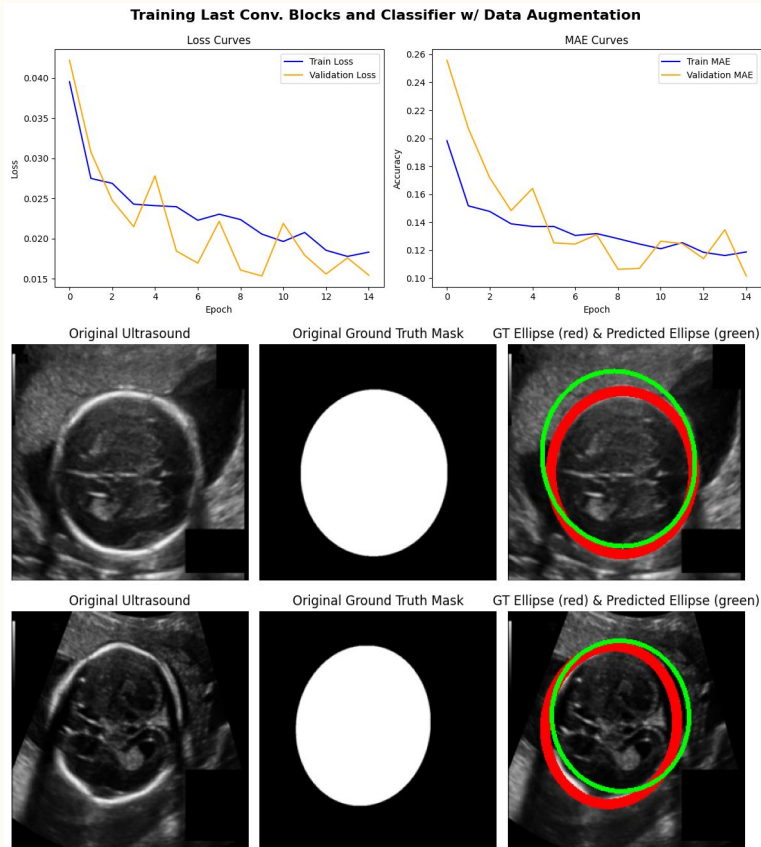


Figure 4. MobileNetV2 + Squeeze-and-Excite [20]. In contrast with [20] we apply the squeeze and excite in the residual layer. We use different nonlinearity depending on the layer, see section 5.2 for details.

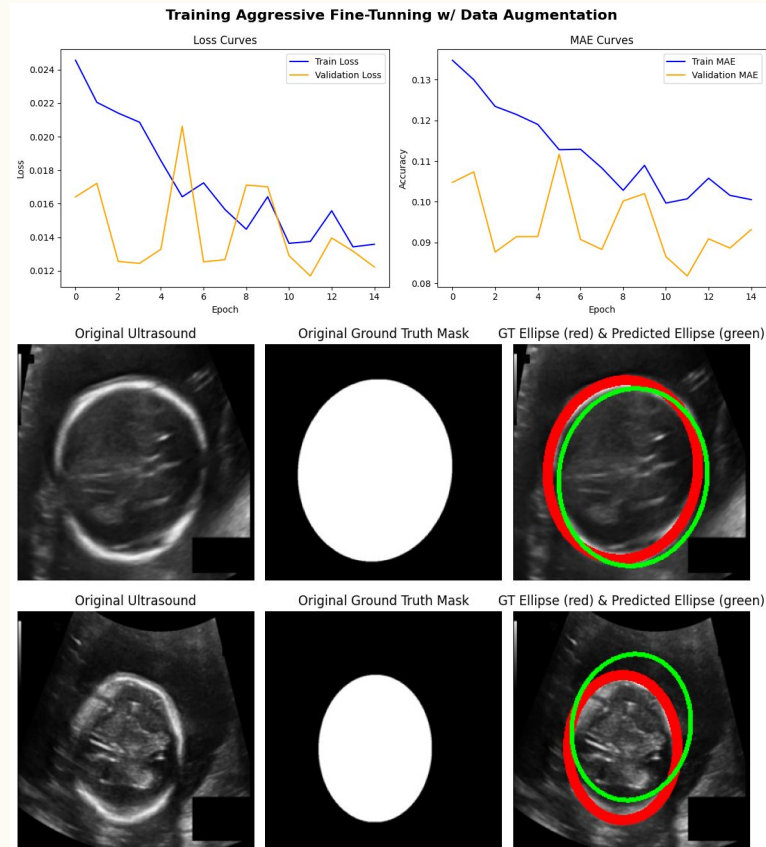
Experimental Results and Analysis MobileNetv3_small



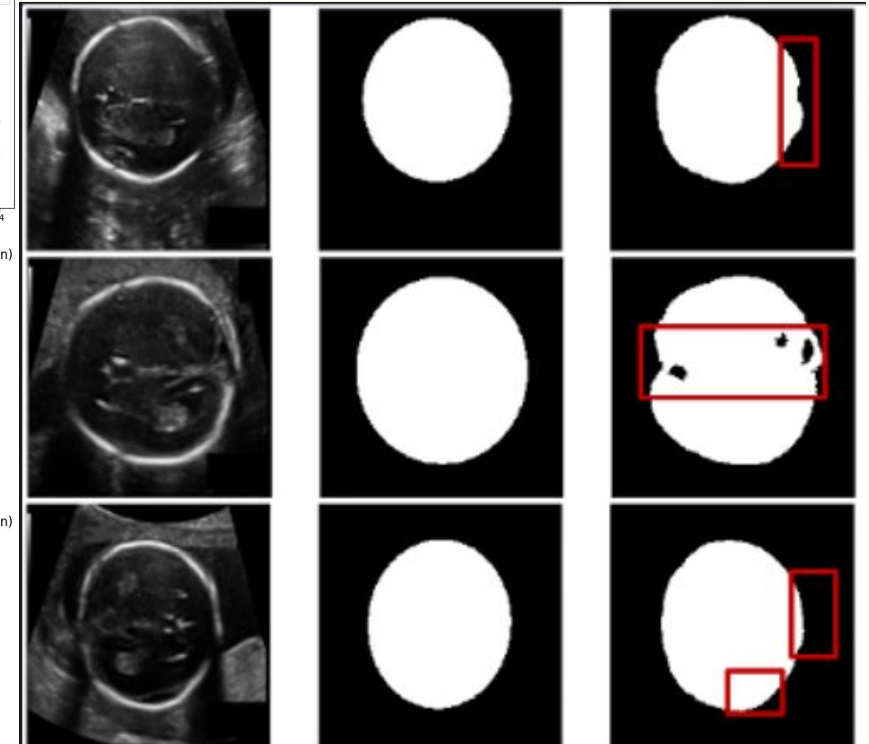
Experimental Results and Analysis Data Augmentation



MobileNetv3_small



MobileNetv3_large



fine-tuned U-Net

Conclusions and Future Work

Successfully implemented a deep learning model capable of automating fetal head measurements

Achievements

Successfully shifted the paradigm from a mask-based segmentation task to a linear classification predicting six key ellipse parameters.

Next Steps

Introduce more complex data augmentation techniques beyond random rotations to better simulate real-world ultrasound noise and further mitigate overfitting.

Thank You for Your Attention

Fetal Head Ultrasound Recognition Project